NEGOTIATING N.Z. SONIC ARTS IN REAL-TIME & NETSPACE

Ian Whalley
Conservatorium of Music
University of Waikato

ABSTRACT

The contention is that it is worthwhile to combine some mainstream historical genres found in New Zealand sonic art – electroacoustic music, traditional Mäori instruments, found/invented instruments, and Asian influenced approaches; and that newer production techniques provide a means of negotiating between different sonic art perspectives in both real-time and netspace. Some early experiments by the author toward this are outlined, particularly in combining tradition Mäori instruments and electroacoustic music. The work *Mittsu no Yugo* (Whalley 2010), a telemusic event between Beijing, Hamilton, and Calgary for MUSICACOUSTICA10, is then discussed as a recent working example of the proposition.

1. INTRODUCTION

The exploration of timbre and gesture in music/sound art to reflect and explore identity has historically been a central concern of musicians/composers. It is then useful to first briefly discuss a theoretical approach to music, genre and identity, to frame why combining some mainstream New Zealand sonic art genres is worthwhile.

In discussing identity generally, Hall ((1996b: 115) notes that individual and group identity is 'formed at the unstable point where the unspeakable stories of subjectivity meet the narratives of history and culture', and it is maintained and updated once established. In this sense, while identity can be fixed, allowing discussion of it, identity is also a narrative used for experimentation or reaffirmation, rather than an attribute of character.

Both Hall (1991: 44-45) and Back (1996: 8) when discussing culture generally, note that people may mix older notions that were once used to define cultural divisions, and that once might have been in opposition, to form new identities. And they also argue that new ethnicities can evolve at a local level as the result of productive tension between global and local influences (Hall 1991: 55, Back 1996).

Music has always played a part in these ongoing processes. A useful bridge as to how this might work sonically is provided by Shepherd and Wicke (1997), who point out that music is a *medium* that allows a dialectic between sounds and people: that music is a process rather than an object. This people/sound dialectic is both structured and structuring in interaction with the human body. Music may then both restrict and facilitate simultaneously the range of meanings that can be constructed by recipients.

As narrative, music as a medium then allows for the creation and construction of experience. Frith (1996: 109) points out that 'music produces people since talking about identity is talking about experience'. Frith (1996: 111) expands on the link between music and identity when he notes that groups do not agree on values and then express these in their cultural activities (a homology model) but that they only get to know themselves *as groups*, *through* cultural activity, through aesthetic judgment. In this sense, making music is not simply a way of representing ideas, but a way of living them.

Given the music/identity link outlined, it is of little surprise that New Zealand sound based artists might want to explore an authentic voice based in their immediate environment (virtual or real) and/or take on aspects of their surrounding cultural influences. Apart from popular music and intermedia forms, four distinct historical New Zealand sonic art approaches might include the following.

The first stems from Douglas Lilburn's (1915-2001) pioneering work. Lilburn abandoned acoustic composition after completing his Third Symphony in 1961, to search for a more distinct New Zealand voice based in the natural environment. Using audio samples fused with electronic sounds made it possible for him to create a unique New Zealand soundscape: part local, part global. Early examples, quoting directly from the environment, are found in his later works like *Three Inscapes* (1972) and *Soundscape with Lake and River* (1979) (see Norris & Young 2001).

The search for authenticity in environment and local culture is also reflected in the emergence of an experimental 'performance' based 'soundculture', resulting from the use of found and invented instruments. Composers such as Phil Dadson, for example, pioneered this approach in New Zealand - his work with *From Scratch*, also incorporated Polynesian rhythms played on tuned percussion instruments made of

plastic pipes. The invented instrument approach extends to sounds sculptures, in works such as Chris Cree Brown's *Aeolian Harp*. (Dart, Elmsly, Whalley, 2001).

As part of an interest in Asian cultures and near neighbours, white New Zealand composers such as Jack Body, for example, took a documentary approach to Indonesian soundscapes in his electroacoustic CD *Suara* (1993) Continuing on from this, the interest in including Asia instruments and perspectives in New Zealand composition is increasingly common across a number musical styles.

But perhaps the most significant recent sonic art development has been the revival of traditional Mäori instruments, using an improvisational approach to creating content, and providing an alternative to Lilburn's sense of New Zealand authenticity. The approach is illustrated in the pioneering work of Hirini Melbourne and Richard Nunns. The CDs *Te Ku Te Whe* (1993) and later *Te Hekenga-ä-rangi* (2003) where a tradition of landscape and language referenced music drawing from a New Zealand pre-European tradition is referenced (see Whalley 2005).

A snapshot of the range of these approaches can be found in *New Zealand Sonic Art III* (Whalley 2002), where the call for works sought material generated from found/invented instruments, traditional instruments, and environmental sounds. The aim was to capture some of the range of cultural approaches and experimental music taking place in New Zealand at the time, to find commonalities and differences between diverse practitioners; and to explore issues of access and elitism, regionalism and internationalism, tools and output. The collection then reflects something of the established but also then emerging and evolving New Zealand identity. A summary of the outcome is given in *New Zealand Electroacoustic Music: Nexus, Flux, Negotiation* (Whalley, 2004a).

2. THINGS TOGETHER, THINGS APART

While each of these distinct sonic art traditions has found ways of expanding their historical base, such as through electroacoustic music's tape and instrument genre, or interaction using programs such as MAX/MSP, the possibilities of bringing together some of these sonic art streams *within works* are yet to be further explored.

There are of course practitioners that have, and continue to, negotiate the boundaries between at least two of the established New Zealand Sonic Art genres, as some of the works on *New Zealand Sonic Art III* illustrate. Further, the crossover between traditional Mäori instrumental approaches and electroacoustic music (see overview: Whalley 2005) is part of a wider interest in combining electroacoustic music with regional instruments/aesthetics internationally (Whalley 2005b, *Organised Sound* issue).

While then appreciating that stylistic diversity is something to be celebrated and expanded; the argument here is that it is often in the synthesis of technical and stylistic approaches that unique hybrids can be found: that music/sound can be used as a way of negotiating between differing perspectives, and that such negotiation is a healthy part of evolving cultural identity.

As part of this process, the development of New Zealand Sonic Art forms is likely to be influenced by the changing environment that we inhabit, one that is increasingly virtual and global through the application of digital technology and is likely to alter our sense of place, space and time. In addition, the increasing use of 'intelligent' agent based machines through automation and decision-making, is likely to influence us, forcing a consideration of the machine role, and 'conscious' machine participation.

Accordingly, concerns with making sound art in real-time, with involving more 'intelligent' machines in the process (Whalley 2009), and with exploring the use of netspace as a performance medium (Weinberg 2005), are areas of interest for computer-based music researchers internationally.

A question when writing the work *Mittsu no Yugo* (Whalley 2010), was then how we might bring at least some of the historical New Zealand Sonic Art traditions together in real-time; and explore new netspace and intelligent machine technology that is now available.

3. THREE TECHNICAL LIMITATIONS AND THREE EARLY QUESTIONS

On the surface, established New Zealand Sonic Art traditions are not immediately suited to our current new real-time digital environment.

Much New Zealand electroacoustic music, while continuing through the application of digital technology and extended through multi-channel diffusion techniques, is still largely based on an editing approach to production. While this is partly mitigated by the 'instrument and tape genre', 'on the fly' digital music generation remains somewhat neglected here. Many of the New Zealand works that use MAX/MSP (see Rowe 1994) as a performance medium in combination with a live instrument at least addresses part of the real-time medium, but without addressing the reactive role of the machine. This is in contrast to Paine's (2002) *conversational model* of interaction as an alternative to machine participation. The question to address here was then how to expand electroacoustic music as a performance idiom to involve the use of machine

intelligence that might improvise in a realistically conversational manner.

The adoption of extensive digital editing techniques to create virtual performances also typified the later work of Nunns and Melbourne. Despite the improvised origins of the material for the second album *Te Hekenga-ä-rangi* made in 2003 for example, digital technology was often used to edit and manipulate sound *after* performances, rather than being integral to performance. As such, the work increasingly drew on electroacoustic music approaches where the result is a hybrid of traditional music and new technology that could not be recreated in live performance. In this sense, *Te Hekenga-ä-rangi* (2003) departs significantly from *Te Ku Te Whe* (1994) in that the production and compositional approach used multiple instrumental parts, and extensive multi-track layering, mixing and effects. The second technical question was then one of how this new hybrid genre might also be realised in real-time.

Finally, the New Zealand 'soundculture' tradition is at least one that has a strong history of real-time composition through improvisation; but in this country, it has not been one that has extensively adopted digital technology as a means of extending sounds, or using effects as a means of manipulating sound after the fact in real-time - particularly in work based on 'instruments' taken from the natural environment. The final question here was then one of how one might extend the idiom through real-time effects processing beyond using simple reactive MAX/MSP patches.

4. SOME EARLY GENRE EXPERIMENTS WITH MÄORI INSTRUMENTS AND EA MUSIC

The central notion of combining New Zealand Sonic Art traditions, and expanding real-time performance with increasing machine intelligence, involved a few byways before arriving at a real-time work in netspace.

My interest in blending different electroacoustic music and Mäori instruments came about after hearing *Te Ku Te Whe,* and then subsequently persuading Melbourne/Nunns to put together a longer and more layered work than appeared on their first album, for *New Zealand Sonic Art III* (2002). Combining improvisation and digital editing, the result was *Te Hau Kuri (Dog's Breath),* the first track on the CD. The short audio sequences for this work were recorded onto a stereo track, with both players improvising using similar instruments panned L/R for most of the sound bites. This provided the source material. Melbourne composed the structure based on a legend he chose from the area he came from. I subsequently produced the track, suggesting further layering to create the sense of a larger ensemble. Apart from pan, EQ, compression and volume alterations that were added to get the ensemble to fit together, the recorded segments are left intact.

Subsequent to this, I took an editing and layering approach to create the work *Kasumi* (Whalley 2002) based on using similar production techniques of gathering short improvised materials to begin. *Kasumi* used samples from seven New Zealand Mäori instruments played by Richard Nunns. However, the source material was also manipulated extensively to create new 'instruments' and sonic textures. In addition, the recorded Mäori instrumental fragments were also loaded into a sampler. Extensive pitch and real-time sound transformation techniques were then added, and the altered sounds blended and contrasted with the original material to create the structure of the work. The sampler also allowed the instruments to be performed in ways that were not physically possible, such as two octave transpositions and extended pitch bending. This overcame the melodic limitations of many of the instruments that are often confined to four notes in one octave. Together with effects manipulations, the result was types of hyper-instruments. Further, the work explored an Asia/Pacific connection integrating Japanese aesthetics. The manipulated Japanese text for this piece is from The Manyoshu, number 4290 by Otomo no Yakamochi (718-785) written on 23 Feburary, 753.

From these first steps, the focus changed to ways in which one might recreate the sense of layering a real-time performance with Mäori instruments, extended through electroacoustic generation and manipulation techniques.

PiWeCS: a Public Space Interactive Web-based Composition System was one of the first outcomes (Whalley 2004b). For this, Richard Nunns provided a range of short samples from a number of instruments. The technical/compositional infrastructure involved creating a web-based system server/client to produce the work. Since access to leading players of the instruments was limited, and many of the instruments unique, the intention was to find ways to overcome barriers of time and scarcity to allow lay people to discover and explore some of the instrumental sounds in a public space, independent of passive reception through live performance and the CD medium; to allow them to explore the combination of some of these instruments in a compositional and interactive and real-time context; and to allow the possibility of extending the instruments in an electroacoustic context as a way of exploring a new hybrid sonic landscape unique to New Zealand.

A further experimental work was the HIEMPA project (Whalley 2008) - Hybrid Instruments from Electroacoustic Manipulation and Models of Pütorino and Aquascape. This project combined a team of people with technical, artistic, environmental and cultural expertise toward an artistic outcome of aiming to extend the New Zealand sonic art tradition. It involved collecting audio samples from the aquascape of the Ruakuri Caves and Nature Reserve in Waitomo, South Waikato, New Zealand, and samples of a variety of

Pütorino – a New Zealand Mäori wind instrument. Following a machine learning analysis of the audio material and an analysis of the performance material, hybrid digital instruments were built and mapped to suitable hardware triggers. The new instruments were playable in real-time, along with the electroacoustic manipulation of Pütorino performances. The project took into account the environmental and cultural significance of the source material, with the results to be released in a series of compositions. Sadly, we were funded for the background research for the project's first stage, but not the production/performance second stage, and so the full CD output remains to be completed. One of the outcomes covering part of the project can be found in *For Morehu Te Whare* (Whalley 2010). In this, two short samples of Mäori musical instruments, the pütorino iti (flute) and the ku (mouth bow) heard in the opening, were played by Richard Nunns. Apart from a short water sample taken from the cave, most of the material is electroacoustically derived and developed.

5. TECHNICAL: 'INTELLIGENT' MACHINES

Perhaps the most useful technical development made as part of the PIWeCS project addressed the need to extend machine intelligence in performance to make something more interactive, in line with Paine's (2002) *conversational model* of interaction. The machine implementation of PIWeCS was achieved through integrating multi-agent programming with MAX/MSP on a server. Human input was through web interface using then FlashMX to build an augmented mixing board. The system developed allowed a limited conversation to be initiated and continued by participants through arranging samples, and building larger compositional structures over time (Whalley 2004b). It also allowed the extension of a composition through the electroacoustic manipulation of source material. The Java based multi-agent system, scripted with compositional knowledge as an 'expert system', acted as a guide and mentor for participants. In addition, the system accumulated user preferences and patterns, allowing the nature of machine/human interaction to be unique to each session. It also allowed the exchange and sharing of ideas and the relationship between users and machine to deepen over time, in line with Paine's model.

6. GENERATIVE IMPROVISATION

The use of intelligent agent systems in music production and performance is of course now widespread and implemented through a variety of aesthetic approaches: a field recently surveyed in *Software agents in music and sound art research/ creative work: current state and a possible direction* (Whalley 2009). In this, I point to how composers, musicians and computer scientists have begun to use agent-based systems to create music and sound art in both linear and non-linear (non-predetermined form and/or content) idioms, with some robust approaches now drawing on various disciplines and industries apart from music. Further, the paper puts forward a theoretical framework for the use of intelligent agent technology in creating music/sound art works, and an overview of common methods of implementation is given.

Software based intelligent agent systems can exhibit varying degrees of persistence, independence, communication and collaboration with other software agents or people. Intelligent agents have the capability to make decisions, the capacity to learn within an environment, and the ability to be mobile over computer networks. More advanced intelligent agents monitor environments, can glean information, make decisions to react or not, and modify their behaviour according to the results received (see Whalley 2009). Software platforms such as *Grasshopper*, *Jade* and *Zeus* for the development of such system, then make the technology well suited to real-time decision making in multi-causal and dynamic situations. A Multi-Agent Systems (MAS) is one where a computational system allows agents to cooperate or compete with others to achieve some individual or collective task(s). A Distributed Artificial Intelligence (DAI) approach is a co-operative system where agents act together to solve a problem, and can include agents being used in decision support systems where the outcome is based on interactions between human and agent input.

Of the four common methods in applying agent based technology to creating form and/or content for sonic art rather than notation-based music, the most prevalent one is perhaps a generative approach reflecting process-based rather than goal-based paradigms of musical creation (Chadabe 1996). A *generative approach* often involves applying biological or evolutionary concepts (also see Dorin 2001), allowing the machine technology to create content and even structure based on the dynamic interplay of parts. This is a non-linear perspective where the composer creates a range of software-agent behaviours, and there will be many possible outcomes and durations of works that result. Process then becomes the artistic focus, rather than a set product. These systems are generally technically self-contained once set in motion, and the language used either self-referential or allowed to evolve (Whalley, 2009). Some recent deployments of intelligent agent technology can develop musical styles, through evolving a machine personality based on internal communication of agents.

But the most significant intelligent agent implementation for human/machine interaction, extending the generative method, lies in a *generative/improvisation approach* to machine agency (Whalley 2009). This allows for real-time improvised human input, but rather than the machine agent simply reacting, there is a human *and* machine adaptation (autonomy and learning) to the generative input of the machine as well as the input of human agency. In theoretical terms, if the adaptive

relationship between human and machine demonstrates balanced listening/dialogue, and if little of the final output is prescribed, a full *conversational model* of interaction results – one where intelligent machines become an equal partner in the creative process.

However, most implementations of the *conversational model* implemented through generative improvisation prescribe some of the machine sonic language in the first instance – an aesthetic judgement to start the creative process. This approach was adopted in the PiWECS system (Whalley 2004b).

Following PiWECS, a further early trial of the generative improvisation method using agent technology was developed with better sensor inputs to capture human gestures, and then applied to make several variations of a soundscape combining Mäori and electroacoustic manipulations. Aspects of the variations were used in the work *Pukera* (Nunns & Whalley 2004). Here, short audio samples were record by Nunns. The sampled fragments appear as recorded at times in the structure, but most are also processed and developed electronically and then blended to make the structure of the work, providing a dramatic dynamic between processed and natural sounds. While the soundscape work is self-contained, for one of the live performances, Nunns used one instrument to improvise a part along with the score. The small Pütorino (wind instrument), played with a variety of techniques, enhanced the texture and melodic shape of the work in real-time (Nunns & Whalley 2004).

Apart from the use of agent-based technology in combination with Mäori instruments, a recent attempt at a generative improvisation approach to making a conversational model of interaction was also explored toward making a real-time model of aspects of Smalley's (1997) view of gesture and spectra. GIIMP - Generative Improvisation and Interactive Music Project (Whalley 2010d) was the outcome. This was realized in MAX/MSP alone, coupled with an additive synthesiser. GIIMP implemented a machine/human generative improvisation system using human gesture input, machine gesture capture, and a gesture mutation module in conjunction with a flocking patch, mapped through microtonal/spectral techniques to sound. The intention of this system was partly to technically simplify some of the previous more complicated agent-based music systems developed, meld some established and current practices, and combine aspects of symbolic and sub-symbolic approaches toward musical outcomes.

7. MITTSU NO YUGO

High-speed Internet2 is being explored as a composition/performance space by an increasing number of electroacoustic music practitioners, and the subject of an upcoming edition of *Organised Sound* (17:1) that I am co-editing. Conceptually, Internet2 technology allows for electroacoustic studios and live performance venues to link across long distances internationally to create new art forms, communicating through data streams as well as high quality audio (Whalley 2010c).

Through the byways of various projects presented in earlier sections, netspace was subsequently explored in the recent international work *Mittsu no Yugo* (12'00) (Whalley 2010b). This was played and 'recomposed' in real-time over high-speed research networks using the new IPv6 format.

The composition was one of a number of ongoing projects with our international partners toward technically exploring sound-based composition/ performance and machine agent approaches suited to Internet2 music making; while also considering and exploring the juxtaposition and synthesis of regional aesthetics in the creation of new compositions.

Mittsu no Yugo as a performance rendition was made in real-time between Hamilton, Beijing and Calgary, and performed live for an audience at the MUSICACOUSTICA10 festival in Beijing.

The Hamilton node had the greatest telepresence in the project - the footprint of the biggest action,. In Hamilton, performers were Ian Whalley (Max/MSP patches, wind synthesizer, pedals), Lara Hall (violin and looper), Hannah Gilmour (spectral beds, rhythm and effects). Richard Nunns' samples were again drawn on and manipulated in the performance. In Canada, David Larsen played Buffalo drum; and in Beijing, Bruce Gremo played shakuhachi.

Aesthetically, the idea was to first explore different locations reflected in distinct instrument timbres and gestures. But the work also attempts to meld aspects of the sonic input from the three different cultural nodes through aspects of real-time conversation and negotiation. As a result, one can sometimes identify individual parts in the work, but sometimes it is difficult to discern who contributed what in the texture. This is partly because many of the real instruments acted as both pitch/duration *and* textural/spectral sound makers; but also because some of the instrumental sonic sources could either be digitally manipulated directly, or manipulated through adding real-time effects.

Net-wise, the technical process involved linking multiple bi-directional audio channels between the three countries, streaming high-quality audio. Independent of this, we linked three video channels using digital video conferencing units. This allowed performers to both see and hear each other live. Apart from the video conferencing units at each node, multiple Skype channels allowed technicians at each station to also type/chat 'off-stage', in case any technical problem arose during the performance.

Small delays with audio latency, shorter than video latency, were compensated for through the way the score was written as a performance guide, allowing for the entry of different parts to be rhythmically imprecise, so that musical timing was not absolutely critical. Accordingly, the work partly adopts a non-realistic paradigm of a Recording Studio/Remixer approach (Carot, Rebelo Renaud: 2007) where latency is accepted and the composition built around its limitations.

In approaching netspace as a composition decision arena from a production perspective, it seems at first blush to be similar to other interactive music decision spaces involving machine technology (Whalley 2009). One still has to make choices between a Process or Goal approach to composition, select some sort of starting sonic language and associated musical/sonic knowledge paradigm, and decide on what the role of the machine is to be in terms of autonomy. But the decision space is complicated by the need to also accommodate some new things that netspace presents.

In theoretically discussing net-based music, Follmer (2005) outlines a three-dimensional schema for NMP (networked music performance). This matrix includes the parameters of Interplay with Network Characteristics, Interactivity and Openness, Complexity and Flexibility. In his subsequent analysis of diverse music practices, Follmer identifies hypermusic (music from hybrid instruments), real-virtual/space sound installations, and algorithmic/generative installations as being best suited to telemusic space. This analysis seemed to resonate well with some of the prior experiences that I have outlined above in developing machine-based real-time electroacoustic music performance systems.

Limiting Follmer's analysis however is its basis in a compositional and machine perspective. Apart from this, a consideration of netspace needs to include communicative issues such as the phenomenology of sound making, music/sound as a means to explore real/virtual space, and social interaction in music making. Complicating this further, players in netspace might consider the phenomenology of wider audience reception, given a worldwide audience through internet broadcast, and they might also consider the reaction of an audience at any real node that takes part in a telemusic event.

To navigate the increasingly multi-dimensional decision space from a compositional perspective, the work on *Mittsu no Yugo* proceeded on the communicative principle of moving from the known to the unknown in sonic terms. This involved combining sonic streams where some parts were familiar to some participants and audience, and some aspects were unfamiliar. The framework for this approach (see Whalley 2010e) considers gesture, sound, language and player semiotics. In the work, it was achieved through layering and balancing three distinct sonic art practices: traditions distinctly associated with the body and 'instrumental'/extensions; environmental and organic sounds and gestures; and abstract gestures and sounds. Combining these approaches resulted in a work that was texturally diverse, yet would have some familiar points of reference for the instrumentalists to start a conversation from, and points of reference for subsequent audiences.

Both Follmer's analysis and the accommodation of a wider communicative dimension lend themselves well to a generative improvisational approach to making aspects of form/content, involving dialogue between players and players/machines. The social aspect of music making between players, for example, involved negotiation and relationship building in real-time that resulted in better sonic outputs as the rehearsals of the work progressed.

From a technical compositional perspective, I began from a goal-based approach in the first instance, because the work had to be played as part of a concert, and this dictated a time length. Accordingly, I wrote a graphic score for the work that loosely prescribed the macro structure, gestures within sections, and broad motives for the work. In addition, I prescribed the sectional tunings and parts of the spectra (see Sethares 2005) that each larger section might adhere to. The key here however, is the flexibility of the structure and the order of any material with each. Participants could make their own decisions within it and the structure could also be changed in real-time in terms of stretching and compressing sections.

Content was then largely made through a process-based approach, and while performers could prepare some of the content they might use, the work was generated interactively in real-time in the dialogue between machine and human participants.



Figure 1. Hall, Gilmour and Whalley at the Waikato node of *Mittsu no Yugo*

As a consequence, the work could arrive at different outcomes with each performance. Moreover, the compositional approach was flexible enough allow for additional performers and instruments to take part and negotiate new outcomes. It is then worthwhile noting that the traditional role of the electroacoustic music composer in netspace starts to shift to one of concept designer, director and participant.

Machine participation, in line with prior work, was first implemented through the real-time manipulation of some of the pre-prepared sample material, allowing for the development of motives using electroacoustic manipulation techniques.

Also adopting previous techniques that had been developed, intelligent machine participation involved a MAX/MSP agent-based system that I built for the Hamilton node. This could generate data and respond to human input, allowing human/machine dialogue in a gesture conversational manner. The output was sonically realised through a combination of resyntheisis and additive synthesis. This sound-making method was used to maintain an organic sensibility that would sometimes integrate with the acoustic instruments being playing in the work, but also contrast with them electroacoustically at times. An area to explore in the future is implementing the system at different nodes, allowing both people/machine dialogue at each node, and machine/machine dialogue between nodes.

Paine's (2002) notion of a conversational model of interaction was then adopted in *Mittsu no Yugo* to include human/machine dialogue. In addition to this, the machine/human relationship also includes reactive machines that some players might use to extend manipulate sounds in real-time or add effects after the fact. But the best implementation of the conversational model however was perhaps implemented in the real-time dialogue between performers as they made the work through netspace.

Work in the medium is ongoing, and an enhanced performance system will be used in an upcoming work between Hamilton, Singapore and Beijing for the upcoming MUSICACOUSTICAII festival.

8. CONCLUSION

The increasing influence of applying digital and Internet2 technology allows composers and musicians to rethink older sonic art forms. As more countries begin to explore the combination of these technologies, we are likely to partly preserve and value the distinct sonic nature of different nations' nodal points on the netspace grid – the sounds and gestures that identify us.

At the same time, as our sense of identity is confronted by others in this new digital space, we are likely to use our established regional sound arts as a means to take part in global conversations, through extending older organic music practices. In tandem, decisions may also need to be made about how to accommodate increasingly intelligent machine technology through intelligent agent technology.

Some interesting research questions then arise. Are we likely to develop broader regionalized aesthetics and undermine national ones? Should machines be subservient to regional aesthetics? And what sort of unique identity might be developed by just machine/machine conversations, or machine/human conversations? Many of these issues are of course far from new, and are likely to have as many answers as there are individual stories that engage with broader narratives of culture and history.

9. REFERENCES

Back, L. 1996. *New Ethnicities and Urban Culture.* London: UCL Press.

Body, J. 1993 *Suara* Ode Record Co. CD Manu 1380.

Carôt, A., Rebelo, P. Renaud, A., 2007. 'Networked Music Performance: State of the Art'. In *Proceedings of the AES 30th International Conference*, Saariselkä, Finland, March 2007.

Chadabe, J. 1996. 'The History of Electronic Music as a Reflection of Structural Paradigms'. *Leonardo Music Journal*, Vol. 6: 41-44.

Dart, W. Elmsly, J. Whalley, I. 2001. 'A view of computer music: Auckland, Waikato and the Asia/Pacific connection'. *Organised Sound* 6(1): 11-20.

Dorin, A. 2001. 'Generative Processes and the Electronic Arts'. *Organised Sound* 6(1): 47–53.

Frith, S. (1996) 'Music and Identity' in, Hall, Stuart & Du Gal, Paul (eds.) *Questions of Cultural Identity.* London: Sage.

Follmer, G. 2005. 'Electronic, social, and aesthetic factors in net music'. *Organized Sound* 10(3), pp 185-192.

Hall, S. 1991. 'Old and New Identities' in *Culture, Gobalization and the World System.* King, Anthony (ed) London: McMillian.

Hall, S. 1996a. 'Introduction: Who Needs Identity' in Hall, S. & Du Gal, P. (eds.) *Questions of Cultural Identity.* London: Sage.

Hall, S. 1996b. 'Minimal Selves, New Ethnicities' in *Black British Cultural Studies: A Reader.* Baker , A. et al (eds). Chicago: University Press.

Melbourne, H., Nunns, R. 1996. CD. *Te Ku Te Whe.* Rattle. RAT D004

Melbourne, H. Nunns, R. 2003. CD *Te Hekenga-ä-rangi.* Rattle. RAT DV010.

Norris, M., Young, J. 2001. 'Half Heard Sounds in the Summer Air: Electroacoustic Music in Wellington and the South Island of New Zealand'. *Organised Sound* 6(1): 121-28.

Nunns, R., Whalley, I 2005. *Pukera* . Organised Sound 10(3) DVD. Track 3

Paine, G. 2002. 'Interactivity, Where to From Here'? *Organised Sound* 7(3): 295-304.

Rowe, R. 1994. *Interactive Music Systems: Machine Listening and Composing.* London: MIT Press.

Shepherd, J. & Wicke, P. 1997 *Music and Cultural Theory.* Cambridge: Polity Press.

Sethares, W. 2005. *Tuning, Timbre, Spectrum, Scale (Second Edition).* London: Springer.

Smalley, D. 1997. 'Spectromorphology: explaining sound-shapes' Organised Sound. **2**(2), pp 107–26.

Weinberg, G. 2005. 'Interconnected Musical Networks: Toward a Theoretical Framework'. *Computer Music Journal* 29:2: 23-39.

Whalley, I (Ed. & Producer). 2002 *New Zealand Sonic Art III.* Hamilton: University of Waikato. CD MDUW1202.

Whalley, I. 2002. Kasumi. *New Zealand Sonic Art III.* University of Waikato, UWMD 1203: Track 5.

Whalley, I. 2004a. 'Recent New Zealand Electroacoustic Music: Nexus, Flux, Negotiation'. *Leonardo Electronic Almanac* Volume 12 (8) August 2004.

Whalley, I. 2004b 'PIWeCS: Enhancing human/machine agency in an interactive composition system'. *Organised Sound* 9(2): 170-176.

Whalley, I. 2005. 'Traditional New Zealand Mäori instruments, composition and digital technology. Some recent collaborations and processes'. *Organised Sound* 10(1): 57-65.

Whalley, I. 2005b. (Ed.) 'New Technology, Non-Western Instruments and Composition'. *Organised Sound* 10(1) pp 1-2.

Whalley, I. 2008. 'HIEMPA: Hybrid Instruments from Electroacoustic Manipulation and Models of Pütorino and Aquascape'. *Organised Sound* 13(3), 259-267.

Whalley, I. 2009. 'Software Agents in Music and Sound Art Research/Creative Work: current state and a possible direction'. *Organised Sound* 14(2): 156–167

Whalley, I. 2010a. *For Morehu Te Whare.* Computer Music Journal (34:4) DVD

Whalley, I. 2010b *Mittsu no Yugo* MUSICACOUSTICA10, Central Conservatory of Music, Beijing, 22-28 October

Whalley, I. 2010c 'Networked electroacoustic music, machines & internet2: A conceptual framework and some artistic issues/responses'. *New Zealand Electroacoustic Music Symposium*, Auckland University 1-3 Sept.

Whalley, I. 2010d **'**Generative Improv. & Interactive Music Project (GIIMP)'. Proceedings of *NIME2010*, June 15-18, 2010, Sydney, Australia, pp 255-258.

Whalley, I. (2010e). 'A Framework for expanding electroacoutic music knowledge'. 7th EMS Network Conference. Shanghai Conservatory of Music. 21- 24 June. pp 109-111.